

AI 101: Beyond the Chatbot

A plain-language field guide for manufacturing, engineering, defense, and public leaders



AI is not magic. It is software that learns patterns from data. A **neural network** learns those patterns from examples. A **large language model (LLM)** predicts useful sequences of words. An **agent** puts a model inside a **harness** so it can pursue a bounded goal.

How we arrived here

Rules › Machine learning › Neural networks › LLMs › Agents

Five building blocks of a modern agent

1 The model (the brain)

Interprets input and proposes the next step.

2 Memory

Supplies approved facts and prior state.

3 Tools (the hands)

Search, calculate, call software, or act.

4 The loop

Observe, plan, act, check, then repeat or stop.

5 Guardrails (the leash)

Limit access, require approvals, log actions, and stop unsafe work.

Harness: the software around the model that connects memory, tools, the loop, and guardrails.

Useful first workflows

- Search approved manuals, procedures, and lessons learned.
- Draft non-final work instructions, reports, or supplier messages.
- Triage maintenance, quality, and supplier signals for expert review.
- Compare requirements while preserving sources and traceability.

Five questions before any pilot

- ☐ **Outcome:** What useful result will we measure?
- ☐ **Boundary:** What may the system see and do?
- ☐ **Evidence:** How will we know when it is wrong?
- ☐ **Authority:** Which decisions stay with a person?
- ☐ **Transition:** How will affected people shape the change?

Take-home exercise: Name one workflow that AI could improve and one boundary it must never cross.

The risks are real and manageable

Challenge	Practical control
Hallucinations	Use approved sources, citations, verification, abstention, and human approval.
Hacking and misuse	Use least privilege, isolation, attack testing, monitoring, logs, and fast shutdown.
Sensitive data	Keep Controlled Unclassified Information (CUI), export-controlled data, and IP in approved environments.
Bias and accountability	Name an owner, test real cases, preserve evidence, and allow appeal.
Job displacement	Map changing tasks, retrain early, redesign roles, and share gains deliberately.
Over-automation	Set action limits, stop conditions, escalation paths, and rollback.

Three things to forbid this week

1. No CUI or export-controlled data in public AI tools.
2. No agent connected to production, enterprise resource planning (ERP), or payment systems without written approval.
3. No AI-drafted compliance or contract submission released without human verification.

CMMC decision rule: AI touching CUI belongs only in Cybersecurity Maturity Model Certification (CMMC)-compliant, contract-approved environments.

For public leaders: Make retraining concrete. Michigan Works! and Macomb Community College are practical local partners for the "where."



Recommended viewing

"Humans Need Not Apply" by CGP Grey (2014). Scan the code or use the link. The video is a useful provocation about automation and work.

<https://youtu.be/CMFj75kBQIU?is=8BTKniNg34Ya80FO>

"We are the horses."

Technology need not replace every human ability to transform the labor market. Prepare people and institutions instead of assuming helplessness.

Discuss: Which of our tasks look most like the horses' jobs, and what would we do about it this quarter?

Humans adapt. Leaders can smooth the transition by protecting what matters, teaching what changes, and shaping AI before disruption shapes them.

AI is moving quickly. What applies today may not apply in a year. Schedule regular refresher courses and update policy and controls. Waiting raises the risk of avoidable job loss and economic recession.

Course materials



Course slides (PDF)



This handout (PDF)

Questions, concerns, or AI policy help?

Reach out to Tristen Pierson, Axiovex Systems.

Tristen Pierson, Axiovex Systems —
Tristen.Pierson@gmail.com —
352-410-0922